RAJALAKSHMI ENGINEERING COLLEGE

Sanjana Banerjee 250701646-2025-CSE S2

| | |
|---|---|
| **Started on** | Friday, 7 August 2026, 8:19 AM |
| **State** | Finished |
| **Completed on** | Friday, 7 August 2026, 8:36 AM |
| **Time taken** | 16 mins 58 secs |
| **Marks** | 1.00/1.00 |
| **Grade** | **10.00** out of 10.00 (**100**%) |

**Question 1** | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using the counter method.

```
void function (int n)
{
    int i= 1;
```

```
    int s =1;
```

```
    while(s <= n)
    {
         i++;
         s += i;
     }
}
```

**Note:** No need of counter increment for declarations and scanf() and count variable printf() statements.

**Input:**

A positive Integer n

**Output:**

Print the value of the counter variable

**For example:**

| Input | Result |
|---|---|
| 9 | 12 |

**Answer:** (penalty regime: 0 %)

Ace editor not ready. Perhaps reload page?

Falling back to raw text area.

```
#include<stdio.h>
int main()
{
    int n;
    scanf("%d",&n);
    int count=0;
    int i=1;
    count++;
    int s=1;
    count++;
    while(s<=n)
    {
        count++;
        i++;
        count++;
        s=s+i;
        count++;
    }
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | 9 | 12 | 12 | ✔ |
| ✔ | 4 | 9 | 9 | ✔ |

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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| | |
|---|---|
| **Started on** | Friday, 7 August 2026, 8:37 AM |
| **State** | Finished |
| **Completed on** | Friday, 7 August 2026, 8:57 AM |
| **Time taken** | 20 mins 8 secs |
| **Marks** | 1.00/1.00 |
| **Grade** | **10.00** out of 10.00 (**100**%) |

**Question 1** | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using the counter method.

```
void func(int n)
{
    if(n==1)
    {
      printf("*");
    }
    else
    {
     for(int i=1; i<=n; i++)
     {
       for(int j=1; j<=n; j++)
       {
          printf("*");
          printf("*");
          break;
       }
     }
   }
 }
```

**Note:** No need of counter increment for declarations and scanf() and count variable printf() statements.

**Input:**

A positive Integer n

**Output:**

Print the value of the counter variable

**Answer:** (penalty regime: 0 %)

```c
#include<stdio.h>
int main()
{
    int n,i,j,count=0;
    scanf("%d",&n);
    count++;
    if(n==1)
    {
        count++;
    }
    else
    {
        count++;
        for(i=1;i<=n;i++)
        {
            count++;
            for(j=1;j<=n;j++)
            {
                count++;
                count++;
                count++;
                count++;
                break;
            }
        }
    }
    printf("%d",count);
    return 0;

}
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | 2 | 12 | 12 | ✔ |
| ✔ | 1000 | 5002 | 5002 | ✔ |
| ✔ | 143 | 717 | 717 | ✔ |

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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| | |
|---|---|
| **Started on** | Friday, 7 August 2026, 9:01 AM |
| **State** | Finished |
| **Completed on** | Friday, 7 August 2026, 9:08 AM |
| **Time taken** | 6 mins 36 secs |
| **Marks** | 1.00/1.00 |
| **Grade** | **10.00** out of 10.00 (**100**%) |

**Question 1** | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using counter method.

```
Factor(num) {
{
   for (i = 1; i <= num;++i)
   {
    if (num % i== 0)
       {
         printf("%d ", i);
       }
    }
 }
```

**Note:** No need of counter increment for declarations and scanf() and counter variable printf() statement.

**Input:**

A positive Integer n

**Output:**

Print the value of the counter variable

**Answer:**

```c
#include<stdio.h>
int main()
{
    int num,i,count=0;
    scanf("%d",&num);
    count++;
    for(i=1;i<=num;i++)
    {
        count++;
        count++;
        if(num%i==0)
        {
            count++;
        }
    }
    printf("%d",count);
    return 0;
}
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | 12 | 31 | 31 | ✔ |
| ✔ | 25 | 54 | 54 | ✔ |
| ✔ | 4 | 12 | 12 | ✔ |

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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| | |
|---|---|
| **Started on** | Friday, 7 August 2026, 9:08 AM |
| **State** | Finished |
| **Completed on** | Friday, 7 August 2026, 9:15 AM |
| **Time taken** | 7 mins 15 secs |
| **Marks** | 1.00/1.00 |
| **Grade** | **10.00** out of 10.00 (**100**%) |

**Question 1** | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time

complexity using counter method.

```
void function(int n)
{
    int c= 0;
    for(int i=n/2; i<n; i++)
        for(int j=1; j<n; j = 2 * j)
            for(int k=1; k<n; k = k * 2)
                c++;
}
```

**Note:** No need of counter increment for declarations and scanf() and count variable printf() statements.

**Input:**

A positive Integer n

**Output:**

Print the value of the counter variable

**Answer:**

```
#include<stdio.h>
int main()
{
    int n,i,j,k,count=0;
    scanf("%d", &n);
    count++;
    for(i =n/2;i<n;i++)
    {
        count++;
        for(j=1;j<n;j=2*j)
        {
            count++;
            for(k=1;k<n;k=k*2)
            {
                count++;
                count++;
            }
            count++;
        }
        count++;
    }
    count++;
    printf("%d",count);
    return 0;
}
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | 4 | 30 | 30 | ✔ |
| ✔ | 10 | 212 | 212 | ✔ |

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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| | |
|---|---|
| **Started on** | Friday, 14 August 2026, 8:13 AM |
| **State** | Finished |
| **Completed on** | Friday, 14 August 2026, 8:18 AM |
| **Time taken** | 4 mins 51 secs |
| **Marks** | 1.00/1.00 |
| **Grade** | **10.00** out of 10.00 (**100**%) |

**Question 1** | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using counter method.

```
void reverse(int n)
{
   int rev = 0, remainder;
   while (n != 0)
    {
        remainder = n % 10;
        rev = rev * 10 + remainder;
        n/= 10;

    }
print(rev);
}
```

**Note:** No need of counter increment for declarations and scanf() and count variable printf() statements.

**Input:**

A positive Integer n

**Output:**

Print the value of the counter variable

**Answer:**

```c
#include <stdio.h>

int main()
{
    int n;
    int rev = 0, remainder;
    int count = 0;

    scanf("%d", &n);

    while (n != 0)
    {
        count++;
        remainder = n % 10;

        count++;
        rev = rev * 10 + remainder;

        count++;
        n /= 10;

        count++;
    }

    count += 3;

    printf("%d", count);

    return 0;
}
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | 12 | 11 | 11 | ✔ |
| ✔ | 1234 | 19 | 19 | ✔ |

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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